

DES Cryptosystem Implementation

Course Title: Information Security Laboratory

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Source Code (DES Cryptosystem):

```
#include <iostream>
#include <string>
#include <vector>
#include <bitset>
using namespace std;
int S[8][4][16] = {
    {{14, 4, 13, 1, 2, 15, 11, 8, 3, 10, 6, 12, 5, 9, 0, 7},
     {0, 15, 7, 4, 14, 2, 13, 1, 10, 6, 12, 11, 9, 5, 3, 8},
     {4, 1, 14, 8, 13, 6, 2, 11, 15, 12, 9, 7, 3, 10, 5, 0},
     {15, 12, 8, 2, 4, 9, 1, 7, 5, 11, 3, 14, 10, 0, 6, 13}},
    {{15, 1, 8, 14, 6, 11, 3, 4, 9, 7, 2, 13, 12, 0, 5, 10},
     {3, 13, 4, 7, 15, 2, 8, 14, 12, 0, 1, 10, 6, 9, 11, 5},
     {0, 14, 7, 11, 10, 4, 13, 1, 5, 8, 12, 6, 9, 3, 2, 15},
     {13, 8, 10, 1, 3, 15, 4, 2, 11, 6, 7, 12, 0, 5, 14, 9}},
    {{10, 0, 9, 14, 6, 3, 15, 5, 1, 13, 12, 7, 11, 4, 2, 8},
     {13, 7, 0, 9, 3, 4, 6, 10, 2, 8, 5, 14, 12, 11, 15, 1},
     {13, 6, 4, 9, 8, 15, 3, 0, 11, 1, 2, 12, 5, 10, 14, 7},
     {1, 10, 13, 0, 6, 9, 8, 7, 4, 15, 14, 3, 11, 5, 2, 12}},
    {{7, 13, 14, 3, 0, 6, 9, 10, 1, 2, 8, 5, 11, 12, 4, 15},
     {13, 8, 11, 5, 6, 15, 0, 3, 4, 7, 2, 12, 1, 10, 14, 9},
     {10, 6, 9, 0, 12, 11, 7, 13, 15, 1, 3, 14, 5, 2, 8, 4},
     {3, 15, 0, 6, 10, 1, 13, 8, 9, 4, 5, 11, 12, 7, 2, 14}},
    {{2, 12, 4, 1, 7, 10, 11, 6, 8, 5, 3, 15, 13, 0, 14, 9},
     {14, 11, 2, 12, 4, 7, 13, 1, 5, 0, 15, 10, 3, 9, 8, 6},
     {4, 2, 1, 11, 10, 13, 7, 8, 15, 9, 12, 5, 6, 3, 0, 14},
     {11, 8, 12, 7, 1, 14, 2, 13, 6, 15, 0, 9, 10, 4, 5, 3}},
    {{12, 1, 10, 15, 9, 2, 6, 8, 0, 13, 3, 4, 14, 7, 5, 11},
     {10, 15, 4, 2, 7, 12, 9, 5, 6, 1, 13, 14, 0, 11, 3, 8},
     {9, 14, 15, 5, 2, 8, 12, 3, 7, 0, 4, 10, 1, 13, 11, 6},
     {4, 3, 2, 12, 9, 5, 15, 10, 11, 14, 1, 7, 6, 0, 8, 13}},
    {{4, 11, 2, 14, 15, 0, 8, 13, 3, 12, 9, 7, 5, 10, 6, 1},
     {13, 0, 11, 7, 4, 9, 1, 10, 14, 3, 5, 12, 2, 15, 8, 6},
     {1, 4, 11, 13, 12, 3, 7, 14, 10, 15, 6, 8, 0, 5, 9, 2},
     {6, 11, 13, 8, 1, 4, 10, 7, 9, 5, 0, 15, 14, 2, 3, 12}},
    {{13, 2, 8, 4, 6, 15, 11, 1, 10, 9, 3, 14, 5, 0, 12, 7},
     {1, 15, 13, 8, 10, 3, 7, 4, 12, 5, 6, 11, 0, 14, 9, 2},
     {7, 11, 4, 1, 9, 12, 14, 2, 0, 6, 10, 13, 15, 3, 5, 8},
     {2, 1, 14, 7, 4, 10, 8, 13, 15, 12, 9, 0, 3, 5, 6, 11}}};
```

```

int EP[8][6] = {
    {31, 0, 1, 2, 3, 4},
    {3, 4, 5, 6, 7, 8},
    {7, 8, 9, 10, 11, 12},
    {11, 12, 13, 14, 15, 16},
    {15, 16, 17, 18, 19, 20},
    {19, 20, 21, 22, 23, 24},
    {23, 24, 25, 26, 27, 28},
    {27, 28, 29, 30, 31, 0}};
int P[4][8] = {
    {15, 6, 19, 20, 28, 11, 27, 16},
    {0, 14, 22, 25, 4, 17, 30, 9},
    {1, 7, 23, 13, 31, 26, 2, 8},
    {18, 12, 29, 5, 21, 10, 3, 24}};
void print(string description, string matrix[], int rows)
{
    cout << description << endl;
    for (int i = 0; i < rows; i++)
    {
        for (auto ch : matrix[i])
            cout << ch << " ";
        cout << endl;
    }
    cout << endl;
}
void des_cryptosystem(string &plaintext, string &key)
{
    string L0 = plaintext.substr(0, 32), R0 = plaintext.substr(32, 32),
           K = key.substr(0, 48);
    string S_[8], ER0[8], K0[8], SB[8], PS[4], R1, L1 = R0, L0_[4], R1_[4];
    int i, j, k;
    cout << "L0: " << L0 << "\nR0: " << R0 << endl;
    for (i = 0; i < 8; i++)
    {
        for (int j = 0; j < 6; j++)
        {
            ER0[i] += R0[EP[i][j]], K0[i] += K[6 * i + j];
            S_[i].push_back(((char)((R0[EP[i][j]] - '0') ^ (K[6 * i + j] - '0'))
+ '0'));
        }
        int j = stoi(string{S_[i][0], S_[i][5]}, nullptr, 2);
        int k = stoi(string{S_[i][1], S_[i][2], S_[i][3], S_[i][4]}, nullptr, 2);
        SB[i] = bitset<4>(S[i][j][k]).to_string();
    }
}

```

```

print("Expanded R0:", ER0, 8), print("K:", K0, 8);
print("XOR of Expanded R0 and K:", S_, 8), print("S Box:", SB, 8);
for (i = 0; i < 4; i++)
{
    for (j = 0; j < 8; j++)
    {
        PS[i].push_back(SB[(P[i][j] / 4)][(P[i][j] % 4)]);
        L0_[i].push_back(L0[i * 8 + j]);
        R1_[i].push_back((char)((((L0[i * 8 + j] - '0') ^ (PS[i][j] - '0'))
+ '0')));
        R1.push_back((char)((((L0[i * 8 + j] - '0') ^ (PS[i][j] - '0')) +
'0')));
    }
}
print("Permutation of S Box: ", PS, 4), print("L0: ", L0_, 4);
print("R1 = XOR of L0 and Permutation of S Box:", R1_, 4);
cout << "L1: " << L1 << "\nR1: " << R1 << "\nC1: " << R1 + L1 << endl;
}
int main()
{
    cout << endl
        << "DES Cryptosystem:-\n";
    string plaintext, key;
    // plaintext =
    "01111001001110101010010010101000001111101011011010010010101";
    // key = "00001011111110110110001101000000001111100110101101011010110";
    cout << "Enter the plaintext(64 bit in binary):";
    cin >> plaintext;
    cout << "Enter the key(64 bit in binary): ";
    cin >> key;
    des_cryptosystem(plaintext, key);
    return 0;
}

```

Input and Output:

```
DES Cryptosystem:-
Enter the plaintext(64 bit in binary):011110010011101010100100101010100000111111010110110010010101
Enter the key(48 bit in binary): 0000101111111011011000110100000000111110011010
L0: 011110010011101010100100101010
R0: 10000011111101011011010010010101
Expanded R0:
1 1 0 0 0 0
0 0 0 1 1 1
1 1 1 1 1 0
1 0 1 0 1 1
1 1 0 1 1 0
1 0 1 0 0 1
0 1 0 0 1 0
1 0 1 0 1 1

K:
0 0 0 0 1 0
1 1 1 1 1 1
1 1 0 1 1 0
1 1 0 0 0 1
1 0 1 0 0 0
0 0 0 0 0 1
1 1 1 1 1 0
0 1 1 0 1 0

XOR of Expanded R0 and K:
1 1 0 0 1 0
1 1 1 0 0 0
0 0 1 0 0 0
0 1 1 0 1 0
0 1 1 1 1 0
1 0 1 0 0 0
1 0 1 1 0 0
1 1 0 0 0 1

S Box:
1 1 0 0
1 0 0 1
0 1 1 0
1 1 0 0
1 0 0 1
0 0 1 0
0 1 1 1
1 1 1 1

Permutation of S Box:
0 0 1 0 1 0 1 1
1 0 1 1 1 0 1 1
1 1 0 1 1 1 0 0
0 1 1 0 0 1 0 0

L0:
0 1 1 1 1 0 0 1
0 0 1 1 1 0 1 0
1 0 1 0 1 0 0 1
0 0 1 0 1 0 1 0

R1 = XOR of L0 and Permutation of S Box:
0 1 0 1 0 0 1 0
1 0 0 0 0 0 0 1
0 1 1 1 0 1 0 1
0 1 0 0 1 1 1 0

L1: 10000011111101011011010010010101
R1: 01010010100000010111010101001110
C1: 01010010100000010111010101011010000011111101011010010010101
```